Deep Learning and Its Applications (CO3133) – Assignment 2

# Deep Learning on Large-Scale Data and Specialized Tasks

## Semantic Segmentation of Urban Street Scenes: A Baseline and a Pretrained Model Compared

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# Problem Formulation

- **Task:** semantic segmentation of urban street scenes, one prediction *per pixel*.
- **Input:** an RGB frame $x \in \mathbb{R}^{3\times H\times W}$.
- **Output:** per-pixel logits $z = f_\theta(x) \in \mathbb{R}^{19\times H\times W}$; the prediction is $\hat{y}_{ij} = \arg\max_c z_{cij}$, so one $512 \times 1024$ frame carries $524{,}288$ predictions.
- **Research question:** with the data, the split and the protocol fixed, how much of the gap between a from-scratch model and a pretrained one comes from *pretrained features*, and how much from architecture, loss or capacity?
- **Dataset:** Cityscapes fine annotations [1].
- **Main difficulty:** the classes span almost three orders of magnitude in pixel share, and the output must keep spatial resolution all the way through.

# Cityscapes

- **Source:** recorded from a vehicle in 27 cities; per-pixel polygons drawn by expert annotators [1].
- **Size:** 5,000 fine frames at $1024 \times 2048$; 11 GB images, 808 MB labels.
- **Licence:** non-commercial research and education, account required.
- **Classes:** 34 label ids, **19 evaluated**; the rest map to an ignore label excluded from loss and metrics.
- **Unlabelled:** 11.8% of pixels, of which 7.7% is the bonnet and rectification border.
- **Bias:** daylight, fair weather, European cities, one camera.




**Figure 1:** Training frames and their label maps.

# Pixel Distribution Decides the Metric

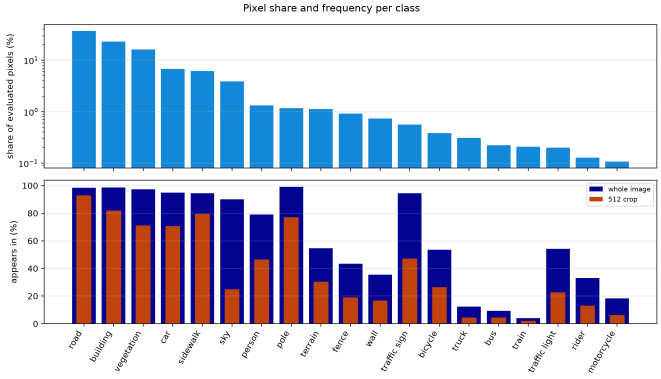



**Figure 2:** Pixel share and frequency per class.

- **Three classes dominate:** road, building and vegetation cover **75.8%** of evaluated pixels.
- Predicting *only* those three already reaches about **0.76 pixel accuracy**.
- **So mIoU is primary** and pixel accuracy decides nothing: mIoU averages over classes, so the other sixteen cannot be hidden.
- Per-class IoU is always reported beside the mean.

# Two Kinds of Scarcity, One Remedy Will Not Fix Both

**Table 1:** Two kinds of scarcity.

| | Class | Pixels | Images | Area when present | Crops |
|---|---|---|---|---|---|
| *Thin* | pole | 1.17% | 99% | 0.83% | 77% |
| | traffic sign | 0.56% | 94% | 0.31% | 47% |
| *Rare but large* | train | 0.21% | **4%** | 0.91% | 2% |
| | bus | 0.22% | **9%** | 0.66% | 4% |

- **Thin classes** are never short of examples: poles appear in 99% of images. They are lost to *boundary imprecision*.
- **Rare-but-large classes** cover 3 to 4× their average share *when present*, but are seldom seen at all.
- **Cropping sharpens the second:** trains fall from 4% of images to 2% of crops. Sky, present in 90% of images, falls to 25% of crops because it sits in a band.
- **A reweighted loss addresses the second, not the first**, which is why the ablation varies the loss.

# Splitting: the Unit Is the City

**Table 2:** Split sizes.

| Ours | Source | Cities | Images |
|---|---|---|---|
| Train | official train | 15 | 2,406 |
| Val | official train | 3 | 569 |
| Test | official val | 3 | 500 |

- **Why the city:** consecutive frames of one street share buildings, parked cars and road surface. Splitting by *frame* would put near-duplicates on both sides.
- **Why not the official test:** its labels are withheld, so the official *validation* partition serves as the test set.
- The three cities that *select* the checkpoint therefore come out of train.
- **Leakage controls:** split committed as a file, not regenerated; normalization never from scored data; test read once, after selection.

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# Pipeline

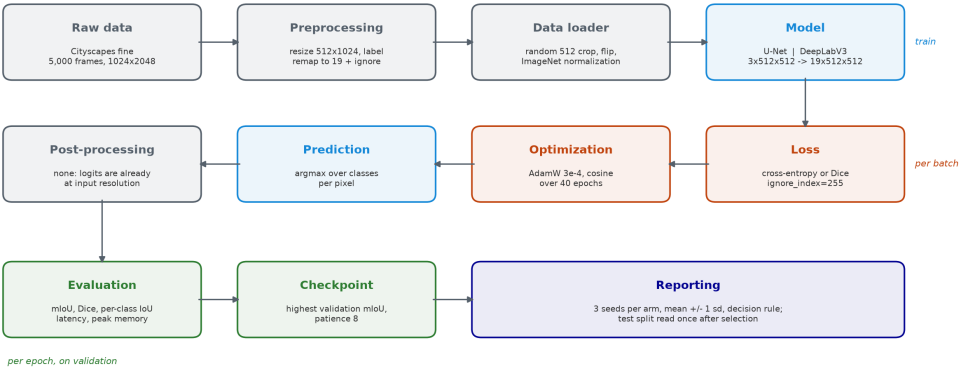



**Figure 3:** The pipeline.

- **Post-processing is empty by design:** both models emit logits at input resolution, so the prediction is an `argmax`; nothing is upsampled or smoothed.
- **Evaluation runs on two terms:** validation every epoch selects the checkpoint; test is read *once*, after selection.

# U-Net, the Baseline

Written from scratch [2]; no pretrained weights. Base width 32, depth 4, $7{,}763{,}635$ parameters.

$$e_\ell = \mathrm{DoubleConv}(\mathrm{MaxPool}_2(e_{\ell-1}))$$
$$d_\ell = \mathrm{DoubleConv}([\,\mathrm{ConvT}_2(d_{\ell+1});\, e_\ell\,])$$

- Widths double, $32 \to 64 \to 128 \to 256$, bridged at 512.
- Convolutions carry **no bias**: each is followed by batch normalisation, which makes it redundant.

**The skip connection is the architecture**

Four poolings take a 512-pixel crop to **32 pixels**, enough context to say *what* a region is, far too coarse to say *where* its boundary runs.

Each decoder stage concatenates the encoder map of its own resolution, putting the upsampled semantics back onto the detail that produced them.

# DeepLabV3, the Pretrained Model

Torchvision DeepLabV3 [3] + MobileNetV3-Large [4], pretrained on a COCO subset over an ImageNet backbone. $11{,}024{,}963$ parameters.

- **No decoder:** dilated kernels keep the feature map at stride 8, and atrous spatial pyramid pooling gathers context at several rates in parallel.
- **Head replaced:** the published $1 \times 1$ convolution predicts 21 classes; ours predicts 19. Backbone and pooling module transfer unchanged.

**Two details that matter**

**Auxiliary head removed**, not computed and discarded, so training pays nothing for an output no loss reads.
**Normalization taken from the weights**, not assumed. Feeding a differently normalised batch to a pretrained backbone degrades it *silently*, with no error to notice.

**Fine-tuning procedure:** one stage. The final convolution is replaced, then *every* parameter trains from epoch 1 at one learning rate, with no gradual unfreezing and no per-layer rates. This is the same procedure the U-Net receives, so the comparison is between *models*, not training schedules.

# Losses and Metrics

**Cross-entropy**, over labelled pixels only:

$$\mathcal{L} = -\frac{1}{|\Omega|}\sum_{(i,j)\in\Omega} \log \frac{\exp(z_{y_{ij}ij})}{\sum_c \exp(z_{cij})}$$

$\Omega = \{(i,j) : y_{ij} \neq 255\}$. Without it, the network would learn to reproduce the bonnet of the recording vehicle.

**Soft Dice** [5], the ablation arm:

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{1}{C}\sum_{c=1}^{C} \frac{2\sum_\Omega p_{cij}\mathbf{1}[y_{ij}{=}c] + \varepsilon}{\sum_\Omega p_{cij} + \sum_\Omega \mathbf{1}[y_{ij}{=}c] + \varepsilon}$$

- **What they weight:** cross-entropy averages over *pixels*, so a class counts in proportion to the ground it covers. Dice averages over *classes*, so a pole counts as much as the road.
- **Metrics:** mIoU primary; Dice, per-class IoU, pixel accuracy secondary. All from one running $19 \times 19$ confusion matrix, because a validation pass holds 298M pixels, so masks are never concatenated.
- mIoU is a `nanmean`: a class absent from the ground truth is *excluded*, not scored zero.

# What Was Written, and What Was Imported

**Table 3:** Attribution.

| Component | Origin |
|---|---|
| U-Net, Dice loss, confusion matrix and every metric | written by the group |
| Training loop, evaluation, experiment runner, figures | written by the group |
| DeepLabV3 and its pretrained weights | `torchvision.models.segmentation` |
| Layer primitives, cross-entropy, AdamW, cosine schedule | PyTorch [6] |
| labelId $\rightarrow$ trainId table | `cityscapesscripts` |
| Plots | matplotlib |

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# One Protocol for Every Arm

**Table 4:** Shared training protocol.

| Setting | Value |
|---|---|
| Input | resize $512 \times 1024$, random $512^2$ crop |
| Optimizer | AdamW [7], $3 \times 10^{-4}$, wd $0.05$ |
| Batch size | 8 (301 steps/epoch) |
| Epochs | 40 max, cosine over 40 |
| Early stop | patience 8 on validation mIoU |
| Checkpoint | highest validation mIoU |
| Seeds | 0, 1, 2 per configuration |
| Hardware | Apple M4 Pro, 14 cores, 48 GB, MPS |
| Software | Python 3.12.0, PyTorch 2.14.0 |
| Precision | full `float32`, no AMP |

## Search effort is zero, and equal

No hyperparameter search was run. Every arm uses the same settings, fixed before training, and changes **exactly one** of them.
Every score is therefore a score *under a common protocol at an untuned configuration*, not the best that configuration could reach.

## Reproducibility limit, reported not hidden

The Apple GPU backend has no deterministic kernel for the scatter-add DeepLabV3 reaches when it interpolates logits back to full resolution. U-Net reproduces bit-exactly; DeepLabV3 diverges by $\sim 4 \times 10^{-7}$ per step.

# Four Experiments, Each Varying One Key

**Table 5:** The experiment matrix.

| Role | Factor varied | Hypothesis | Se |
|---|---|---|---|
| Main comparison | U-Net from scratch vs. pretrained DeepLabV3 | pretrained wins at equal capacity | 3 |
| **Controlled experiment** | backbone frozen vs. fully fine-tuned | full fine-tune wins; widest on thin classes | |
| **Ablation** | cross-entropy vs. Dice loss | Dice lifts train, bus, truck | |
| Extension | MobileNetV3 vs. ResNet50 backbone | ResNet50 wins at $3.6\times$ params | |

**Decision rule, fixed before any result was seen**

$A$ beats $B$ *only* when its mean validation mIoU is higher **and** the two intervals of mean $\pm$ one standard deviation across three seeds **do not overlap**.

Overlapping intervals $\Rightarrow$ **indistinguishable**, which is itself a result and is *not* resolved by picking the higher mean. A claim about one class applies the same rule to that class's IoU. One seed supports no claim of superiority.

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# Main Comparison

**Table 6:** Main comparison.

| Configuration | Params | s/ep | h | ms@1 | GB | Val mIoU | Test mIoU |
|---|---|---|---|---|---|---|---|
| DeepLabV3-ResNet50 | 39,638,355 | 739.9 | 8.22 | 72.3 | 4.37 | *0.6665* | *0.7076* |
| **DeepLabV3-MobileNetV3** | 11,024,963 | 140.4 | 1.52 | 15.2 | 3.41 | $\mathbf{0.6136 \pm 0.0051}$ | $\mathbf{0.6180 \pm 0.0049}$ |
| frozen backbone | 8,053,011 | 95.3 | 0.84 | 14.7 | 3.41 | $0.4961 \pm 0.0049$ | $0.5003 \pm 0.0088$ |
| Dice loss | 11,024,963 | 150.5 | 0.50 | 14.9 | 3.41 | $0.4958 \pm 0.0045$ | $0.4983 \pm 0.0123$ |
| U-Net, from scratch | 7,763,635 | 260.2 | 2.89 | 22.6 | 6.57 | $0.4836 \pm 0.0154$ | $0.4702 \pm 0.0071$ |

- **One large effect, three ways of losing it.** Only the fully fine-tuned pretrained model separates. The other three three-seed arms are **mutually indistinguishable**, within 0.0126 of each other; the frozen and Dice arms differ by **0.0003**.
- **Pixel accuracy ranks them the other way round:** U-Net 0.9202 beats frozen 0.8959 and Dice 0.8982, while losing on mIoU. Exactly what the EDA predicted.

# Decision Rule Applied

**Table 7:** Every pairwise comparison, on validation mIoU.

| Comparison | Verdict | Margin |
|---|---|---|
| DeepLabV3 vs. U-Net | DeepLabV3 better | 0.1301 |
| DeepLabV3 vs. frozen backbone | Full fine-tuning better | 0.1175 |
| DeepLabV3 vs. Dice loss | Cross-entropy better | 0.1178 |
| U-Net vs. frozen backbone | **Indistinguishable** | 0.0126 |
| U-Net vs. Dice loss | **Indistinguishable** | 0.0123 |
| Frozen vs. Dice loss | **Indistinguishable** | 0.0003 |
| Anything vs. ResNet50 | Indicative only, one seed | n/a |

**Why three seeds were necessary**

The U-Net's seed spread is $0.0154$, **three times** that of the pretrained arms, with seeds at $0.4687$, $0.4824$, $0.4995$. Its *best* seed would have tied the frozen arm; its worst would have lost clearly. A one-seed study could have reported either outcome.

# Training Behaviour

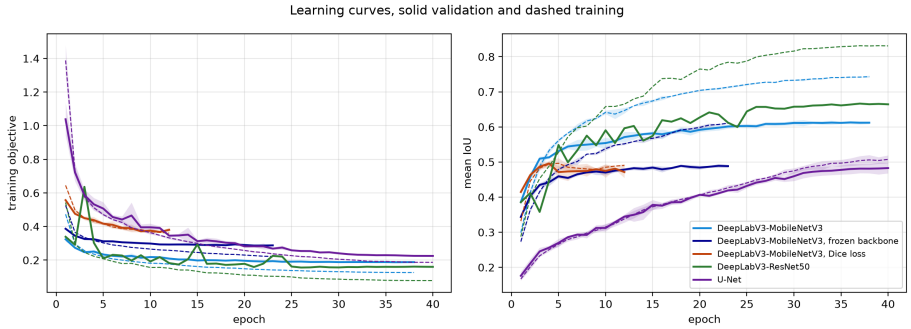



**Figure 4:** Solid validation, dashed training; the left axis mixes two objectives.

- **Budget was right for DeepLabV3:** best epoch 30/31/35 of 38/39/40, flat from about epoch 32.
- **The U-Net is undertrained:** best epoch 38/40/39 of 40, still rising at the last epoch. **0.4836 is a floor**, so the margin over it is an upper bound.
- **Dice peaked at epoch 4** on all three seeds and stopped at 12.

# Where the Difference Appears

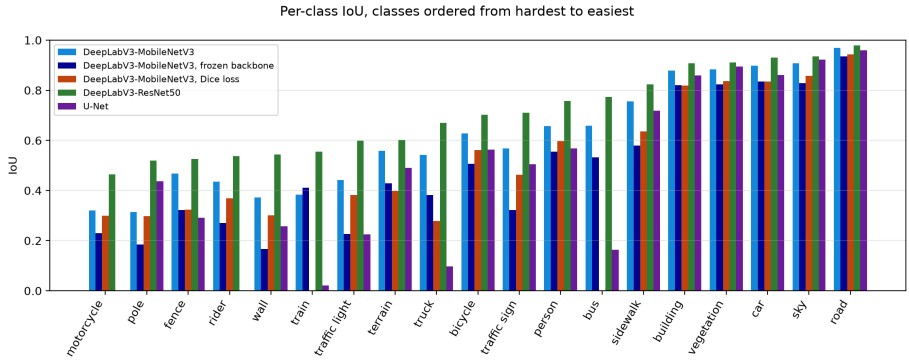



**Figure 5:** Per-class IoU on test, hardest class first.

- **Frozen loses on all 19 classes**, but very unevenly: 0.027 on road, 0.044 on car against 0.215 on truck, 0.213 on traffic sign and 0.173 on traffic light. A frozen backbone keeps what fills a street scene and loses what does not.
- **Pretraining is not uniformly better.** The from-scratch U-Net *beats* DeepLabV3, with separated intervals, on **pole** (0.4126 vs. 0.3277), sky and vegetation.

# Compute Cost

## The U-Net is Pareto-dominated

30% *fewer* parameters than DeepLabV3, yet

- $1.9\times$ the seconds per epoch,
- $1.5\times$ the latency at batch 1, $2.7\times$ at batch 8,
- $1.9\times$ the peak memory (6.57 vs. 3.41 GB),
- and $0.130$ lower mIoU.

Parameter count is **not** a proxy for cost: its convolutions run at full resolution and its decoder holds every encoder map alive.

## Freezing is a training-only saving

$0.84$ h against $1.52$ h, a 45% saving, for $0.118$ mIoU.
Smaller than it sounds: **73% of parameters stay trainable**, because the ASPP head holds 8.05M against the backbone's 2.97M.
**Inference cost is identical** (14.7 vs. 15.2 ms), because the same forward pass runs either way.

## Scaling the backbone

ResNet50: $+0.0529$ val mIoU for $5.3\times$ s/epoch and $4.8\times$ latency. One ResNet50 run costs more than the nine MobileNetV3 runs combined.

# Qualitative Results

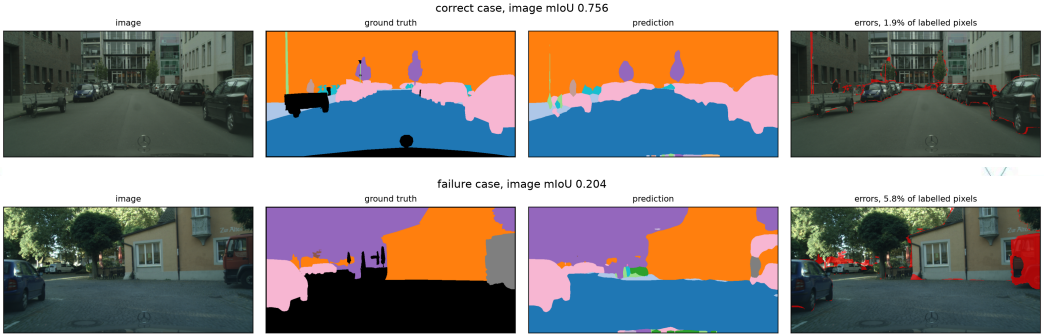



**Figure 6:** Best image (0.756 mIoU) above, worst (0.204) below.

- **Correct case:** the error is a thin outline tracing every boundary, so the model knows *what*, and is imprecise about *where*.
- **The failure case has only 5.8% of labelled pixels wrong**, *below* the 6.7% split average, yet is the worst image in the split. Most of the courtyard carries the ignore label, so few pixels are scored and each class has tiny support. **Per-image mIoU and per-image error are only loosely coupled.**

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# Error Taxonomy

**Table 8:** Error taxonomy, as a share of each model's own errors.

| Group | DeepLabV3 | ResNet50 | Frozen | Dice | U-Net |
|---|---|---|---|---|---|
| Thin structures | 24.7% | 23.0% | 19.6% | 21.9% | 18.7% |
| Upright surfaces | 24.0% | 26.5% | 24.2% | 24.0% | 23.4% |
| Participant vs. background | 17.5% | 15.7% | 16.5% | 17.7% | 15.8% |
| Ground surfaces | 14.6% | 14.8% | 20.3% | 17.4% | 17.1% |
| Traffic participants | 6.5% | 6.8% | 6.3% | 6.4% | **12.5%** |
| Other | 12.7% | 13.3% | 13.1% | 12.7% | 12.4% |
| **Error rate** | 6.7% | **4.9%** | 10.4% | 10.2% | 8.0% |

**The error structure is a property of the task, not of the model**

Error rates differ by more than $2\times$ across the five configurations, yet the composition is nearly identical. Improving the model lowers the total and leaves the proportions alone.

# The Groups, and What Causes Them

- **Thin structures, 24.7%.** Pole $\rightarrow$ building is the single largest flow in the matrix (0.91M pixels, 5.9% of all errors); pole $\rightarrow$ vegetation adds 3.6%. Pole is the hardest of the 19 classes (0.3230 IoU) *despite appearing in 99% of images*, so this is a **resolution failure, not data scarcity**. A stride-8 feature map upsampled bilinearly cannot represent a structure a few pixels wide.
- **Upright surfaces, 24.0%.** Building $\leftrightarrow$ vegetation, 1.16M and 1.13M pixels, *nearly symmetric*, which is the signature of a boundary error rather than a bias. Wall and fence add a taxonomy problem: low barriers separated from building by convention, not appearance.
- **Ground surfaces, 14.6%.** The road/sidewalk kerb is a few pixels high and often occluded by parked cars. Grows to 20.3% for the frozen arm.
- **Traffic participants, 6.5%, and the one place a model differs in *kind*.** 12.5% for the U-Net, nearly double. It scores **0.0205 on rider** against DeepLabV3's 0.4316. Rider and person differ only in what the person sits on, which requires recognising the vehicle beneath. **This is where pretrained features help most.**

**What would improve it**

Five of six groups are dominated by *boundary* pixels $\Rightarrow$ the lever is **resolution**, not the objective. Train and evaluate at native $1024 \times 2048$; add a full-resolution decoder path. **Not** a reweighted loss: the ablation measures that directly, and it makes things worse.

# Ablation: Cross-Entropy vs. Dice Loss

**Hypothesis (pre-registered):** Dice raises IoU on train, bus and truck, which are rare but large when present, because it weights classes rather than pixels.

**Table 9:** The ablation, on validation IoU.

| Class | Cross-entropy | Dice | Change | P(absent) |
|---|---|---|---|---|
| train | $0.6007 \pm 0.0383$ | $\mathbf{0.0000 \pm 0.0000}$ | $-0.6007$ | 85.1% |
| bus | $0.4154 \pm 0.0923$ | $0.1944 \pm 0.1696$ | $-0.2211$ | 70.0% |
| truck | $0.3797 \pm 0.0398$ | $0.1952 \pm 0.0233$ | $-0.1845$ | 69.3% |
| motorcycle | $0.4661 \pm 0.0064$ | $0.2775 \pm 0.0313$ | $-0.1886$ | 60.2% |
| traffic sign | $0.5518 \pm 0.0036$ | $0.4692 \pm 0.0068$ | $-0.0826$ | 0.6% |
| pole | $0.3277 \pm 0.0085$ | $0.3048 \pm 0.0072$ | $-0.0229$ | 0.0% |
| road | $0.9722 \pm 0.0004$ | $0.9502 \pm 0.0016$ | $-0.0220$ | 0.0% |

**Refuted, and in the opposite direction**

Dice is worse on *all nineteen* classes. Correlation between P(absent) and the damage Dice does: $\mathbf{r = -0.809}$.

# Why Dice Fails Here

## The mechanism

For a class with *no* positive pixels in the batch:

- **Cross-entropy** gives it no term of its own; it is only a competitor in the softmax denominator.
- **Dice** gives it a full term $\dfrac{0+\varepsilon}{\sum_{\Omega} p_{cij} + 0 + \varepsilon}$, maximised by driving that class's probability to **zero everywhere**.

Dice **actively suppresses** every class missing from the current eight crops.

## Measured

$10.43$ of 19 classes appear in a given $512^2$ crop; $15.92$ in a batch of 8.
So **16.2%** of Dice terms in a typical step are suppression terms.
For *train*, present in 2% of crops: roughly **85% of the 301 steps per epoch push its logit down**, about 45 push it up.
$\Rightarrow$ the only exact zero in this report: under Dice the model never predicts a single train pixel, on any seed, on either split.

**It also explains the curve:** peaked at epoch 4 on all three seeds, train-to-validation gap $-0.002$. Not overfitting, not underfitting: the suppression terms balance the overlap terms and the objective stops producing useful gradient.
**What this does *not* establish:** that class balancing is useless here. Accumulating Dice over an epoch, or masking absent classes out of the mean, avoids this exact mechanism. Neither was run.

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# Answering the Research Question

**How much of the gap between a from-scratch model and a pretrained one comes from pretrained features rather than architecture, loss or capacity?**

**A sharp answer**

Fully fine-tuned pretrained features are worth

- $\mathbf{0.1301}$ validation mIoU over the from-scratch U-Net,
- $\mathbf{0.1175}$ over the *same weights* with the backbone frozen,
- $\mathbf{0.1178}$ over the *same model* trained with a class-balanced objective,

and those three alternatives are **mutually indistinguishable**, within 0.0126 of one another. One factor accounts for essentially the whole spread.

**An unexpected qualification**

The advantage is **not uniform**. The from-scratch U-Net is better, with separated intervals, on **pole, sky and vegetation**, because a full-resolution decoder localises thin structures that a stride-8 backbone cannot. Five of six error groups are boundary-dominated.
**Pretraining supplies recognition rather than resolution.**

Carried forward: **DeepLabV3-MobileNetV3, fully fine-tuned**, at $\mathbf{0.6180 \pm 0.0049}$ test mIoU and $0.7441 \pm 0.0048$ test Dice, 140 s/epoch, 15.2 ms/image.

# Limitations

- **The U-Net is undertrained.** Validation mIoU still rising at epoch 40 on all three seeds, so $0.4836$ is a floor and the margin over it an *upper bound*.
- **The model comparison confounds two factors.** Its arms differ in architecture *and* in pretraining; no run here separates them. A DeepLabV3 trained from scratch would.
- **The backbone extension is one seed** and supports no claim of superiority.
- **Three seeds estimate a standard deviation from three observations.** "Indistinguishable" means *not separated at this sample size*, not equal.
- **The data is narrow**: daylight, fair weather, European cities, one camera, and images were resized to a quarter of their pixels for compute, itself a ceiling on the thin classes that dominate the errors.
- **Reproducibility has a hardware limit.** The Apple GPU backend has no deterministic kernel for the interpolation DeepLabV3 uses, so rerunning a DeepLabV3 seed will not reproduce its number exactly. The reported spread includes that nondeterminism.

# What Comes Next

In the order the evidence supports them:

1. **Resolution.** Train and evaluate at the native $1024 \times 2048$, or add a full-resolution decoder path (DeepLabV3+). This attacks the thin-structure and boundary groups, which are most of the error, and the $512 \times 1024$ resize was a choice made for compute rather than accuracy.
2. **The clean version of the model comparison.** A DeepLabV3 trained from scratch separates pretraining from architecture in a single run.
3. **The class-balancing question the ablation *reopened*.** Dice accumulated over an epoch, or masking absent classes out of its mean, avoids the suppression mechanism measured here.

**Two practices that changed the conclusions**

Reporting per-class numbers beside the mean, and fixing the decision rule before seeing results. The per-class table located the one setting in which the *weaker* model wins, and the rule turned three apparently different outcomes into a single tie.

# Thank You

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